# Serve: technical report

**Play more. Manage less.** Meta Global AI Developer Hackathon 2026 · Indie Developer track · Build for Your World

Serve is a self-evolving sports social platform built to beat the inertia that keeps people from playing. Its agent does the hassle (it finds the court and the people, agrees the match with the other players' agents, and asks the player only for a yes), and what the community asks for makes the platform better through a safety review and changes that roll themselves back. It is built from verifiable agents: code computes the facts and keeps the rules, the models choose only where a real choice is left, and every text a model writes is checked before anyone sees it (3.8).

**What Serve is not.** It is not a coach or a sports analyzer: it gives no technique feedback and scores no one's performance. It is not a medical tool: it gives no diagnosis, treatment or dosage. What a player says about their body is a scheduling constraint (doubles instead of singles, a shorter game, one game fewer), and a highlight is a clip to share with friends.

Every number below names the file it comes from. Timings name the hardware: one NVIDIA RTX 3090 on my GPU server, or cloud A100s serving the same Muse Glimmer build (the same model digest), reached through that server. The product runs on the RTX 3090 I own; the cloud A100s ran evaluations only.

**What running Muse Glimmer taught me first.** It reasons before it answers, even when told not to. Over the 2,755 calls in the evaluations' ledgers (the student's 4,868 training choices were a separate run, 3.12), 84% of the tokens it wrote were reasoning and 9% of calls came back empty; 222 of those 248 empty replies were the quick pick, whose small budget ran out while it was still reasoning. With thinking switched off it still reasoned in a hidden channel and returned nothing. So every call asks for `think: "low"` and has room to reason beyond its answer (900 tokens by default), and each cheaper tier was measured against that (`submission/results/glimmer_in_practice.json`; 5.9).

**Built during the hackathon.** Everything in the repository was written during the building period (October 27 to November 4, 2026). The one earlier piece of mine is TrackNetV5, a published shuttle tracker, fetched at run time and off unless `BALL_TRACKING=1`; none of its code is in the repository. The code is at https://github.com/gino6178/serve-meta-hackathon, restructured into its current layout on October 28 and 29.

**Live with invited players (since 29 October).** Serve runs on a public address for invited players (3.17). By the afternoon of 29 October: 8 people (14 accounts), 2 clubs players started themselves, 4 games posted, 7 joins and 2 games filled, besides the demo club (see 7). Each player's agent runs in a process of its own on my server.

**Simulated people are labelled as simulated wherever they appear.** The demo club and the larger clubs used for the evaluations are generated by code. None of their members is a real person.

## 1. What Serve does for a player

Most people who stop playing a sport have not stopped liking it. Inertia wins, because finding a court, finding people and matching up is a hassle: a time that suits four people, a free court, someone to push them. In a 2015 poll of 2,506 US adults by NPR, the Robert Wood Johnson Foundation and the Harvard Chan School, 40% of 18 to 21 year olds played a sport and 26% of 26 to 49 year olds [32]. Asked what stops them doing more, adults in Sport New Zealand's Active NZ survey (2023) put other commitments first (64.2%), then getting out of the habit (21.3%); 10.2% have no one to do it with [33]. Those are the three things Serve's agent takes on: it fits the game around the week, asks before the habit lapses, and finds the people. The aim is less time on the phone and more on court: the work happens between agents, and what reaches the player is one question ("Thursday 19:00: Leo is free. Want a game?") and one answer ("Game on: Thursday 19:00, 4 in"). Nothing in it is tied to one sport: any game that needs people, a time and a place works the same way. Serve rests on two ideas.

**Your agent does the hassle,** with three helpers:

- **Match** watches the player's usual playing times. When one is open in the next three days and the people they play best with are free, it asks first: "Thursday 19:00: Leo is free. Want a game?"
- **Talk** takes the yes to the other players' agents and agrees the time and the court.
- **Body** keeps a body log and reads the player's health-app export, and bends the plan around them: doubles while a knee is sore, one game fewer in a heavy week, a check-in two days later.

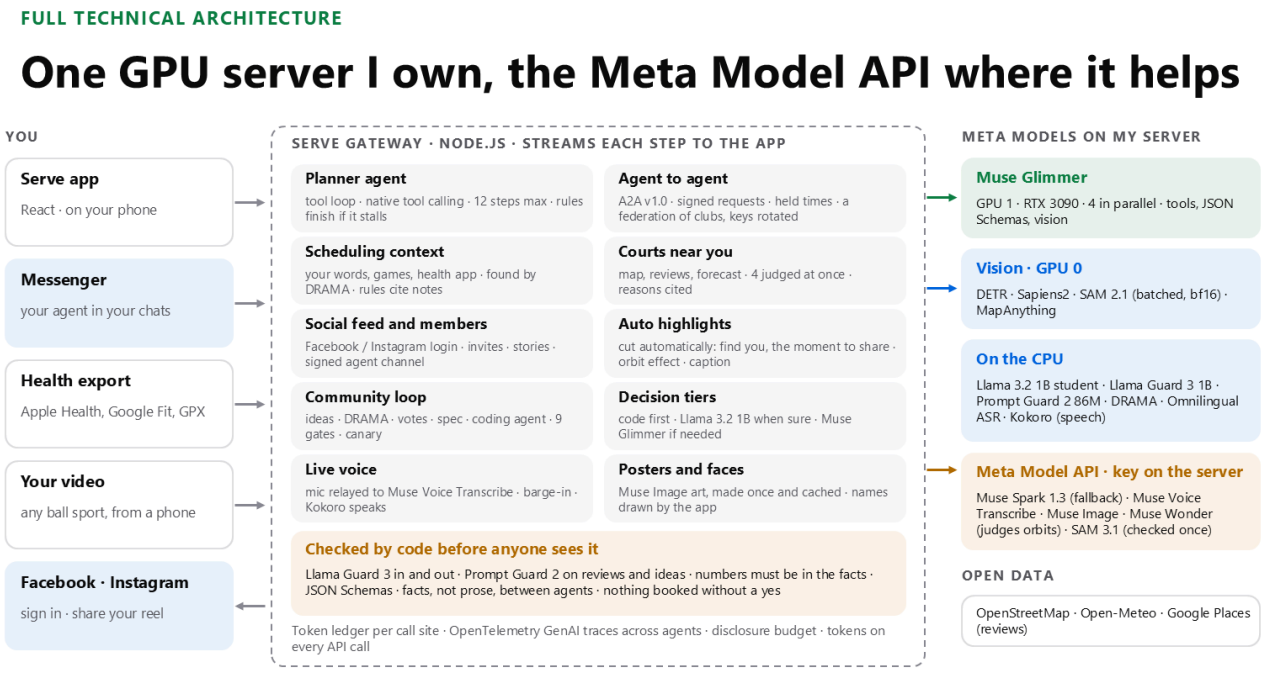
The **Autopilot** plans a whole week against a goal. "I want to play" in Messenger starts **matchmaking**: the agent asks the other members' agents how likely a yes is and talks to the likely ones first. Nothing is booked without the player's yes.

**The community makes the platform better** (3.7). A member tells their agent what they wish Serve did; with their yes, the wish is shared as one checked sentence. Shared wishes are screened, grouped and voted on, become proposals, and pass a safety review. A safe setting ships behind a flag to a canary of members and turns itself off when it does harm; anything else waits for a person, and a coding agent builds it on a branch that only a person merges. This loop is measured on a simulated club (5.6), not yet on a real one.

Around these sit a club (feed, games anyone can join, challenges, a random match, a best match), courts near you in any ball sport, highlight clips from the player's own video, voice notes, and the same agent in Messenger.

**Who else it serves.** Games are hardest to start where no one has the time to organise them. A club on Serve can be one of three kinds for those groups (3.19): children's sessions a teacher or volunteer runs, gentle games for older adults, and a charity tournament.

## 2. Architecture




Architecture

- **One gateway.** A Node.js gateway (`agent/src/server.js`) runs every agent on my GPU server and streams each step to the React web app as NDJSON, so the player watches the agent work.
- **Where the models run.** Muse Glimmer: GPU 1, an RTX 3090, under Ollama 0.40.11 with a 16,384-token context and 4 requests in parallel (`submission/results/manifest.json`). DETR, Sapiens2, SAM 2.1 and MapAnything (for orbit replays): GPU 0. Llama Guard 3 1B, Prompt Guard 2 86M, DRAMA and Omnilingual ASR: the server's CPU, so they take no GPU memory from Glimmer.

- **One client for every model call** (`agent/src/core/llm.js`). It keeps to the model server's 4 slots, puts calls a player is waiting on ahead of background work, asks for JSON Schemas, and logs every call by call site in a token ledger (`ledger.js`).
- **Traces.** OpenTelemetry spans named by the GenAI semantic conventions [20]. The trace context travels inside each A2A message, so one negotiation across all its agents is one trace.
- **Code layout.** `agent/src/` has one folder per concern: `core/` (the model client, ledger, traces, schemas, the player store, auth), `safety/` (the guards, the goal check, privacy, disclosure), `agents/` (the answer, the Autopilot, "want a game?", the decider), `a2a/`, `match/`, `loop/` (the community loop), `feed/`, and `club/`, `body/`, `memory/`, `media/`, `channels/`. `server.js` is the entry point; the web app is in `web/`.
- **Members' agents** are A2A agents with signed Agent Cards, served by the same gateway. **Messenger** reaches it through a signed webhook.

## 3. The agent systems

One split holds throughout. Code computes the facts (times, courts, trips, rules, counts) and makes every decision the facts can settle. Muse Glimmer chooses where a real choice remains and writes short text, which code checks first.

### 3.1 Match and the Autopilot

**Match** (`agent/src/agents/ask.js`) runs once a day, at the hour in the `ask.hour` flag (default 12:00). Code looks `ask.ahead` days ahead (default 3) for a usual time still open, applies the body log, ranks partners by past games that ran past their booking, and finds a free court. Glimmer words the question from those facts.

**The Autopilot** (`agent/src/agents/autopilot.js`) is Glimmer in a tool loop with native tool calling, in the ReAct pattern [1]: `get_week`, `check_limits`, `find_partners`, `find_places`, `negotiate`, `request_approval`, `nudge`, `recall`, `finish` (`check_limits` was `check_body` in the earlier logs). Code runs each tool with checked arguments and returns facts; a refused call comes back as a fact and the model re-plans. After a step budget (12), rules finish the plan with the same limits. A stop check (`notYet`) refuses an early finish, at most twice a run, while an open time still has players free who were not asked. It came from an earlier run that stopped a three-game week at two (`submission/agent_eval.md`).

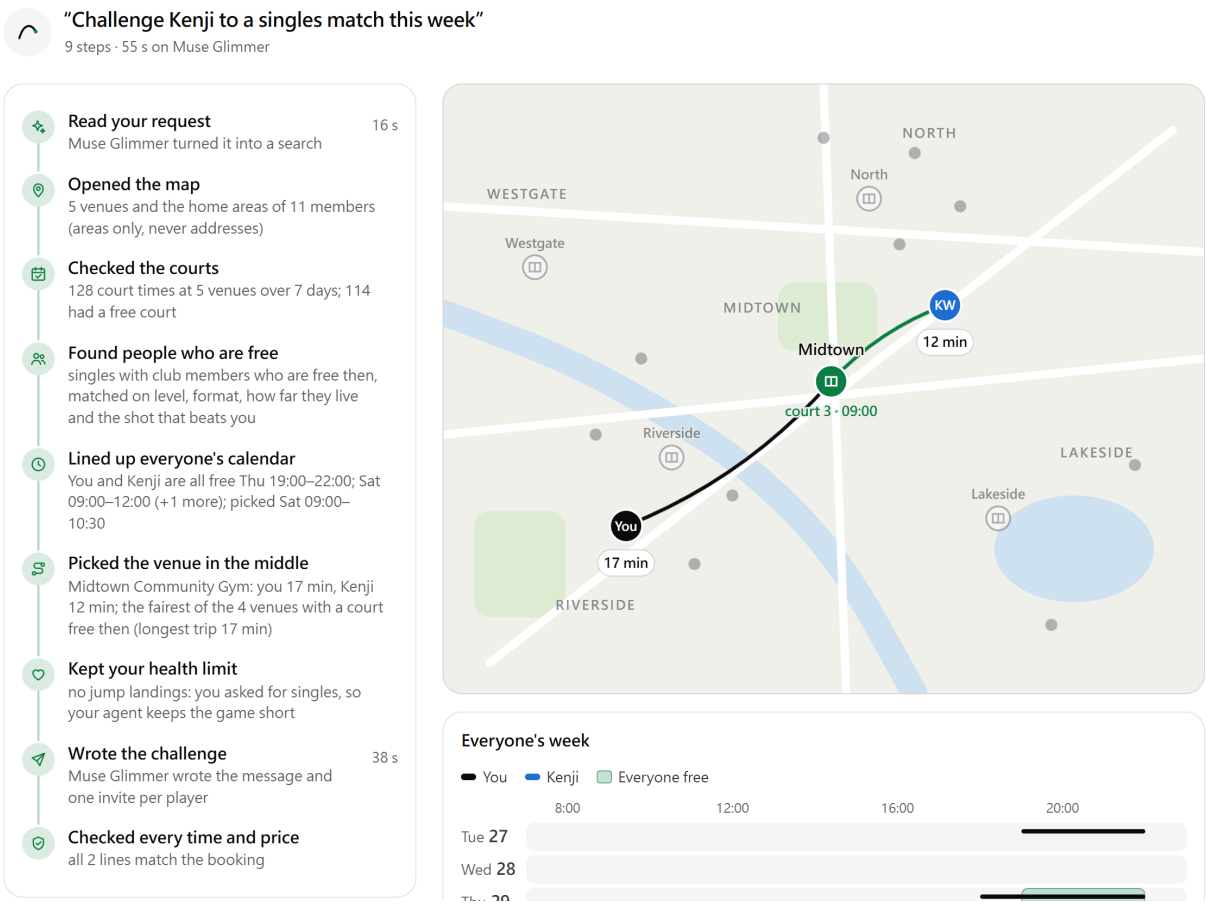
### 3.2 Agent to agent over A2A

`agent/src/a2a/talk.js`, `server.js`, `client.js`, on the official A2A JavaScript SDK. A2A v1.0 is a Linux Foundation protocol [19], not Meta's.

- **Facts only.** A proposal carries each time as `{id, date, start, venue, court, trip}` and nothing else; a decision carries `{intent, date, start, works}`. Sentences between agents are templates filled by code, so neither side's model reads the other side's words.
- **Acceptable sets.** Your agent lists up to twelve times that fit every calendar and have a court. Each member's agent marks which work, and that is binding. Your agent takes the best time in the intersection, so most talks end in round one: the Contract Net [2] with a one-shot intersection, a simple distributed constraint satisfaction [3]. A study of agent markets found that the models it tested favoured the first offer they saw [30]. Here that can't happen: the agents only mark which times work, and code picks from the intersection.
- **Hard rules in code.** A member's rules (days off, the latest finish, the longest trip) take times off the list before the model sees it.
- **Routing by facts** (`A2A_ROUTE`, on by default). For a member whose limits are all rules in code (10 of the 11 demo members), a time that keeps them is a yes and a single remaining time is the counter, with no model call.
- **Holds.** An agreed time is held by every member's agent for 30 minutes, all or nothing, members asked in a fixed order: leases [4] with resource ordering [5]. The booking checks every partner again [6].

- **Durable and signed.** A talk is saved before every round, a message id is answered once, and booking is idempotent. Agent Cards carry a JWS over their JCS form (Ed25519) [7, 8], checked before any talk.
- **Private calendars** (`A2A_PRIVATE=1` sets the server's default, off unless set; each player's own switch in the app wins). Your agent proposes from your own week, each member's agent checks its own calendar in code, and ruled-out times are not proposed again, for three rounds at most.

**Where to play: the venue in the middle.** Each candidate time comes with a place, chosen from the map by code (`agent/src/a2a/talk.js` `court`). For every venue with a free court at that time, code works out each player's trip from their home area (areas only, never addresses) and keeps the venue whose longest trip is shortest. That is the fairest place for the group, not the nearest one for the asker. The trips travel with the proposal, so a member's own longest-trip rule can refuse a place that is too far for them, and the plan shows the route and each person's minutes on the club map. For sports the club's venues don't offer, the agent looks for public courts from OpenStreetMap near the player instead (3.3).




The venue in the middle: each player's trip on the club map, and the step that picked it

## 3.3 Scheduling limits (body log, health import) and courts

What a player says about their body becomes a scheduling limit, never advice. Glimmer files it with one tool call (part, side, level, cause), checked by code (`agent/src/body/body.js`). A health export (Apple Health, Google Fit, GPX, TCX or CSV) is parsed on the server with Python's standard library and deleted (`health_import.py`). Code turns them into scheduling limits (doubles first, a shorter game, one game fewer). It records and schedules; it gives no diagnosis, treatment or dosage.

Courts (`agent/src/club/places.js`) come from OpenStreetMap, with Google Maps ratings and reviews when the server has a key and the Open-Meteo forecast. Glimmer judges four places at once; a reason that does not cite the map, a review (word for word) or the forecast is dropped. Prompt Guard 2 reads the reviews first.

### 3.4 Automatic highlight editing, and orbit replays

A highlight is cut automatically from the player's own video: a clip to share with friends, made with no editing by the player. Nothing in it scores technique or performance. In `video/pose/reel.py`, DETR [16] finds the people in the player's own video, Sapiens2 finds the moment worth sharing and SAM 2.1 [17] cuts the player out. The clip replays that moment in slow motion, with a caption Glimmer writes and code checks.

`video/pose/orbit.py` is the app's optional camera effect ("Add a camera effect"): time stops and the camera moves round the player. MapAnything [18] reads metric depth and the camera from the moment's frame, SAM 2.1 cuts the player out, the court behind them comes from neighbouring frames, and a learned threshold on invented pixels picks the widest arc that still looks right, or a zoom. `video/pose/exists.py` decides frame by frame whether the player and the racket hand are there, from SAM 2.1 as a video tracker, the DETR track and Sapiens2's confidence, with hysteresis.




Time stopped, the camera 12 degrees to the left

### 3.5 Memory by meaning

What a player tells the agent in Messenger is kept as short notes on the server and embedded with DRAMA [9] (`facebook/drama-base`, 768 dimensions, CPU), loaded as a plain Llama with bidirectional attention so no code from its repository runs (`agent/src/memory/memory.js`, `embed_worker.py`). The closest notes come with the Autopilot's week, `recall` fetches more, and someone the player asked to leave out is kept out by code. Retrieval is scored on its own, before any model reads the notes, as LongMemEval does [10].

### 3.6 Guard models

Llama Guard 3 1B [11] runs on its own CPU-only Ollama (`agent/src/safety/guard.js`). It reads what the player writes in Messenger (waiting at most 4 s) and model text for the player (at most 8 s). Prompt Guard 2 86M reads Google reviews before the court judge (`guard_worker.py`). Two rules came out of measuring:

- Llama Guard flagged ordinary injury advice as S6 often enough that it reached the player about seven times in ten. S6 now blocks a text only when code also finds a dose, a medicine to take, or a condition the player is said to have.

- Code also reads plain words of self-harm, which get a helpline reply even if Llama Guard misses them or is down.

In chat a guard that fails is skipped and counted. In the community loop's screen it fails closed.

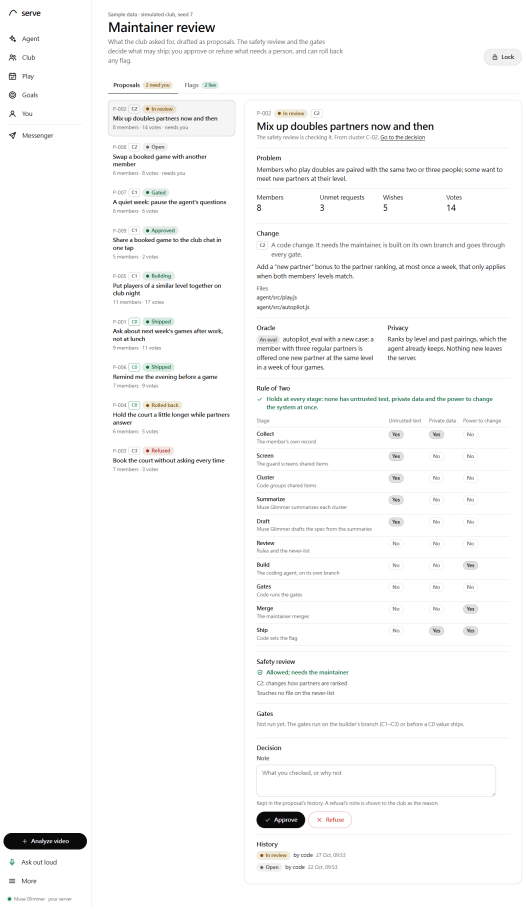
### 3.7 The community self-improvement loop

What the club says becomes proposals; a safe change ships behind a flag to a canary and turns itself off when it does harm (`agent/src/loop/`). Code makes every decision, models write short sentences that code checks, and only a C0 change (a declared setting, inside its range) ships without a person.

1. **Signals.** With `/improve on`, a member's agent counts thumbs, unmet requests and answers to "want a game?", keyed by an HMAC of the player id, with no words. Body and health never become signals.
2. **One item, one yes.** Glimmer paraphrases an unmet request as one sentence; code checks it (no names, numbers, links or body terms), and the member taps Share or Don't.
3. **Screen.** Prompt Guard 2, Llama Guard 3 and code again; a required guard that is down holds the item.
4. **Ideas.** DRAMA clusters the sentences. An idea shows at 5 distinct members and becomes a proposal only with weekday and weekend support. Votes are quadratic: 9 credits a month, n votes cost $n^2$ [12].
5. **Spec.** Glimmer summarizes a cluster (no tools, nothing private), then drafts a spec from the summary alone. Code writes the counts, the class, the oracle, the privacy line and the Rule-of-Two rows.
6. **Review** (`loop/review.js`, `test/policy.mjs`). The class comes from what the change does (C0 a setting, C1 wording, C2 a tool parameter, C3 anything else), never from the draft. The never-list (`docs/never.md`: booking without a yes, data off the server, a diagnosis or dose, weakening `/forget`, editing the tests or the guard, and more) is checked on the spec and any diff, with patterns that follow CodeShield's rules from LlamaFirewall [13].
7. **Gates** (`loop/gates.js`): policy, code scan, unit tests, crash matrix, red team, Messenger end to end, the Autopilot at pass^3 = 1.0, the oracle, a privacy canary. They stop at the first failure.
8. **Ship and watch.** A C0 change goes to beta members in an HMAC bucket under 20%, for 7 days. A hard breach (a booking without a yes, a double booking, the privacy marker leaving) turns the flag off at once. A soft breach (thumbs down up 10 points with a one-sided two-proportion z of 1.645 or more, or requests not falling) turns it off and reopens the proposal.

A C1 to C3 change waits for a maintainer. After their yes, **the builder** (`loop/builder.js`), Glimmer in a tool loop, writes it on a branch in its own git worktree. It reads only the spec's change, files and oracle, never community text; tests and the never-list are read-only to it; its tests run with no network; a maintainer merges.

This draws on the Darwin Gödel Machine [14] (which hacked its own evaluation, so tests are read-only here), AlphaEvolve [15] (keep a change only when evaluators pass it), Clio [21] (summaries and minimum cluster sizes for privacy) and CaMeL [22] (the model that reads untrusted text is not the code that acts).




Maintainer review on the web app's sample data: class, Rule-of-Two rows, decision

### 3.8 Checked before anyone sees it

Numbers in a model's text must be in the facts it was given; a quoted review must be in that review; days and times in a summary must belong to an agreed game; no "he" or "she". Most failing texts go to a template; the body report is redrafted once, then replaced by rules.

### 3.9 Matchmaking: whose agent to ask first

`agent/src/match/`, `docs/agent_matchmaking_plan.md`. "I want to play" (or "anyone for doubles Thursday?") in Messenger becomes an intent: Muse Glimmer reads it into a JSON Schema (`play.js` `readIntent`), after Llama Guard.

- **The ask.** Your agent sends `serve.likely` to up to twelve candidate members' agents at once (ten by fit, two drawn from the rest). It is a new A2A message, declared as an extension in each signed Agent Card.
- **The answer stays coarse.** Each member's agent works out the chance of a yes from its own member's records (how they answered asks lately, how often and when they play, games with this asker, and how often this asker's yeses became games) and returns only a band: unlikely, maybe, likely or very likely. The records never leave it. The model is logistic regression with isotonic calibration [27], trained on simulated logs only.
- **Opt-in.** Simulated and demo members answer by default. A real player's agent answers nothing until the player says `/bands on` in Messenger or flips the switch on the You page. A member may also choose to share the exact chance, rounded to 0.05 and only from 0.15 up, so it reveals their week no more than a band does.
- **Against probing and inflation.** Refusals all read the same; 10 estimates a week per asker and member and 40 a week per member; a 30-minute cache; an audit; and a calibration ledger on the asking side that catches agents whose bands run high. An unknown band is ordered as the club's average, never as a no. `/forget` deletes the records and sets every choice back to off.
- **Fair exposure** [26], a club setting, on by default. Your agent orders by band, then by who was asked least this week, then by fit, so the same few members are not asked every time.

- **Then the usual talk.** The ordered members go through the same A2A negotiation as 3.2. The agreed game comes back as a question with Book it and Drop it buttons; nothing is booked before the tap. For a game today, the player may choose to ask two members at once.
- **Retraining** on consenting members' records (`/learn on`) runs on the server's CPU and installs a new model only if it passes a calibration gate on held-out records; the old model is archived and can be rolled back.

The same order feeds the Autopilot's `find_partners`, the Match helper's daily ask and the Play page.

### 3.10 The club feed: agents and people post together

The Club page's feed (`agent/src/feed/`, `docs/club_feed.md`) is written by agents and people side by side, and every card says which. The club's reading agent finds recent sports-science papers on OpenAlex, arXiv and Semantic Scholar, deduplicates them by DOI, arXiv id and title, and posts two or three plain sentences on each. It never shows the abstract. Prompt Guard 2 reads the abstract first, Muse Glimmer writes, code checks the text, and Llama Guard 3 reads it on the way out. How many papers a week is a flag of the community loop (`feed.papers.perWeek`, 0 to 7, default 3). A member's agent drafts a highlight from the member's own video, and a booked game becomes a one-line recap. The highlight goes up only on that member's yes. The recap goes up only when every player in it says yes: one no, or no answer within 48 hours, and it is never posted. People post notes and photos, and like, comment, share and report. Muse Glimmer looks at each photo under a strict policy (JSON Schema verdict) before anyone sees it, and every check fails closed. A question in a comment on a paper is answered by the asker's own agent from the abstract, with the link. The maintainer works through reports on `#/maintain`, and every action goes into an audit log.

### 3.11 A2A between two gateways

Until this step every member's agent ran inside one gateway. A second, independent Serve gateway now runs on another machine (`agent/src/a2a/peer.js`, `docs/a2a_two_gateways.md`), with ten simulated members of its own, and your agent negotiates with them over the network: proposals, holds, bookings and "how likely is a yes" all cross between the machines. Each gateway trusts the other only as far as its signed requests, its pinned key and its allow-list go. No port is open to the internet. Both listen on loopback, and an SSH tunnel with a pinned host key is the only way between them. In the measured run (12 concurrent talks across the two machines, 22 of 22 checks), the hop to the second gateway, on a GCP A100 in us-central1, added about 185 ms per request. Every time B's agents agreed to was inside their own calendars and rules, no member was booked twice, and a booking whose time another talk had taken in the meantime was refused by B's agent and left nothing behind on either side.

### 3.12 A distilled decision model in front of Glimmer

When a member's agent answers a proposal, code decides everything the facts decide. What reaches a model is one judgment: which of several workable times the member would like best. A student model answers that without writing any text (`docs/decision_model.md`). It is Llama 3.2 1B Instruct, distilled from Muse Glimmer's own answers, quantized to int4 and served on the CPU, and it returns a probability for each declared option. It answers only when it is sure (0.6) and its reading of the facts matches code's, and only for members whose limits are all in code. Otherwise Glimmer answers, also when the worker is down or slow. Its options are only times that keep every hard rule, so it can pick a time the member likes less, but never one that breaks a rule. In talks with the simulated club it cut Glimmer's member answers by 54 to 67% with no rule broken. On 500 held-out cases it agrees with Glimmer 89.5% of the time, against Glimmer's 88.8% agreement with itself. It was trained on 4,868 choices Glimmer made through the real agent call, a separate run on a cloud A100 (`docs/decision_model.md`). On an idle 8-thread CPU the 1B answers in about 0.9 s (about 2 s on the server at low load), against about 40 s for Glimmer's full answer. At the served threshold (0.6) it answers 73% of the 196 hardest cases and 96% of the 500 test cases; Glimmer answers the rest. If the student is busy for 8 s or has not answered in 20 s, Glimmer answers. As served, it keeps the member's stated preference in 29 of 32 cases (Glimmer alone 27).

### 3.13 What talks reveal, and a disclosure budget

Private calendars keep each member's week out of the talk, but every talk still answers yes or no for each listed time. I measured what a curious agent can rebuild from those answers over a month (`docs/privacy_budget.md`, `submission/results/privacy_probe.json`), following CalBench's setup. Without a defence, five well-chosen talks pin 97.6% of a target's weekdays exactly. A budget on hours alone does little until it is small enough to cost ordinary players games. What works is a check on the member's side (`agent/src/safety/disclosure.js`): an answer that would make a whole day certain to that asker is withheld. It brings the days known down to 32.4%, and the ordinary player still agrees every game (its best time drops from 88.9% to 81.5%). It cannot hide the hours a game needs: to agree a game the asker must learn that some hours are free.

### 3.14 A federation of clubs

The two-gateway A2A link (3.11) is generalized to any number of clubs (`agent/src/a2a/peer.js`). A club publishes a card, `/a2a/club.json`: its name, the members other clubs may ask (made-up demo members only, never a real player), a public profile each, and where its keys are. A gateway's `A2A_FEDERATION_FILE` lists only addresses; it reads each card at start and every ten minutes, and each club becomes a peer whose key is pinned on first use. Keys can be rotated: the old key signs a hand-over statement (`typ: serve-key-rotation+jwt`) naming the new key, published in the key set next to it. When a request arrives under a key it does not know, a gateway follows the statements from the key it pinned, and re-pins only if every link verifies; a statement signed by any other key, or a page that is not a card, is refused. `test/federation_unit.mjs` (4 of 4) runs three clubs over real HTTP: discovery, signed requests and a replay refused, a rotation followed two statements deep, and a forged rotation refused with the pin unchanged.

### 3.15 Private calendars, all the way through

The talk between agents could already keep calendars private (3.11). Now the code that plans for one player does too (`agent/src/agents/freebusy.js`): with private calendars on, the Match helper's daily ask, the planner's `find_partners` and its check before a talk, the play search and best match never read another member's free times. They ask that member's agent "free on this date at this time for 90 minutes?" and get a yes or a no. One asker may ask about one member at most 24 different times a day (`A2A_PROBE_BUDGET`), so a week cannot be read back hour by hour; past the budget the answer is no and the refusal is counted. The calendar the player sees shows the partners' rows as the one time their agents said yes to. `test/freebusy_unit.mjs` seals each member's calendar so that any read outside its own agent throws: the play search and the daily ask find the same best time and partners as with shared calendars, and the shared mode is shown to read them directly.

### 3.16 A learned intent router behind the expressions

A game request in Messenger is recognised by regular expressions (`match/flow.js isPlayRequest`). A message they miss now goes to a small softmax head on DRAMA vectors (`agent/src/match/router.js`, trained on the CPU in numpy by `test/router_train.py`; five classes: play, played, body, question, chat). When "play" is the most likely class the message goes to matchmaking, whose game still waits for the player's yes. Trained on 132 messages: 111 the simulated users wrote in the three runs after the fixes and 21 phrasings written for the expressions' own tests; tested on the judge's held-out set, written separately in a person's style (32 messages in these classes). The expressions found 6 of 10 game requests; the router 9 of 10; the two together 10 of 10 (`submission/results/intent_router.json`).

### 3.17 Live: an agent of their own for each player, clubs on a real map, invites from a sentence

**One address, an agent each** (`agent/src/tenants/proxy.js`, `ops/tenants.sh`). A Cloudflare tunnel publishes a small proxy on the GPU server. It serves the web app's public build (no gateway token in it: the browser holds only the member's session) and routes by path: sign-in, members, clubs, the feed and A2A go to the club's gateway, which

everyone shares; every other call goes to the signed-in member's own gateway, a separate process started on their first call, with its own data folder (memory, body log, plans, prefs) and stopped after 20 idle minutes. Leaving deletes the member's folder with their account. A restart of the proxy takes the running gateways up again.

**Clubs anyone can start** (`agent/src/members/clubs.js`, `web/src/pages/Clubs.jsx`). A member starts a club with a name, a sport and a place picked on a real map (OpenStreetMap through Leaflet; the courts nearby from OpenStreetMap, or Google Places when OpenStreetMap's public servers are busy). Public clubs: anyone sees their games and posts, and posting joins. Private clubs: on the map, but their posts are for members, who ask to join and the owner says yes. The name and line are checked like any post. A member's agent recommends the nearest public club within 15 km.

**Invites from a sentence** (`play.js inviteDraft`, `web/src/pages/Messages.jsx`). Told "Saturday morning doubles, need 3" (or 「我想打羽球」), the agent drafts the invite: the sport and format, the day worked out by code from the words, the time, how many players, and the nearest court to the player. The player changes the count and taps Post; it goes into their club for that sport, else the shared feed, and the chat shows who has joined.

**An invite as a link** (`web/src/pages/InviteLink.jsx`, `GET /api/invite`). Any invite can be shared from the feed or the chat (the phone's share sheet, else the link copied) to a group chat or to someone not on Serve. The link opens a card anyone can read without an account: the sport, the time, the place on the map and the spots left, never who is going, and nothing from a private club. Signing in from it (a demo account, Facebook or Instagram) joins the game straight away, also after a login that left the page. That is how a new club starts with no one on Serve yet.

### 3.18 Four pieces of agent architecture

**Active epistemic querying, by the value of information** (`agent/src/match/voi.js`). When a game request's talks have not landed, the agent does not give up or ask the player again. For each member and each of the player's times it keeps a Beta posterior of "free then", Beta(1 + yes, 1 + no), over what it has seen of that member's agent at that weekday and hour (the faded counts of 3.2, never a calendar), and multiplies it by the chance of a yes from the member's band. It then asks the one yes-or-no question whose answer most raises the expected best chance of a game, through the private free-or-busy channel and its budget (3.15), and talks again only on a yes. On 20,000 simulated requests (8 members, 6 times, hidden true availability; `test/voi_sim.mjs`, `submission/results/voi_sim.json`), one question by value of information then one talk found a game 51.9% of the time, against 40.7% with no question, 44.4% with a random question and 48.2% asking about the likeliest pair.

**A critic in the loop on every counter-offer** (`agent/src/a2a/critic.js`). Before the player's agent may take a member's agent's counter-offer, code reviews it for the player. Probing: a member's agent that asks for more than two different times in one talk is mapping the player's week, and its further counters are refused. Fairness: the trips of the games the agent agreed are kept; when the player had the longest trip in two of their last three games, a counter that again gives them the longest trip by more than 10 minutes is refused, and the talk goes on with the other times. Each refusal is said in the trail, with the reason.

**From episodes to beliefs** (`agent/src/memory/beliefs.js`). Once a week (or on ask), Muse Glimmer abstracts the player's live notes into a few standing beliefs about how they like to play, each citing the notes it rests on. Code checks every belief: the cited notes exist and are live, a belief that generalizes ("usually", "prefers") cites at least two, and every number in it is in what it cites; a belief that fails is dropped. The planner reads the beliefs first, each with how many notes stand behind it; the notes stay, and `/forget` deletes both. On 10 written notes, Glimmer gave 3 beliefs, each resting on two notes, none dropped, in 42 s (`submission/results/beliefs_glimmer.json`), for example "work-night games start no earlier than 19:00 and finish by 22:00".

**Proof of consent before a booking** (`agent/src/a2a/consent.js`). When a talk agrees a game, each member's agent that said yes signs the game's canonical record (date, start, length, venue, sport and the sorted players) with its own Ed25519 key; the player's agent signs when the player says yes. Booking verifies every signature against

the game it is asked to book: a record changed after it was signed (another time or venue, a player added) or a signature made with another agent's key is refused. Members on a peer gateway confirm through their own gateway's mandate (3.11). `CONSENT=require` also refuses a game that carries no signatures.

The four are tested without a model in `test/agentic_unit.mjs`; the offline suites now hold 275 tests (25 suites, with `test/programs_unit.mjs` and the invite tests in `test/feed_unit.mjs`).

### 3.19 Clubs for groups with no organiser

`agent/src/members/programs.js` and `clubs.js`, `web/src/club/Programs.jsx`. A member who starts a club picks its kind. Every rule below is code; no model decides who plays when.

- **Children's sessions**, for a teacher or volunteer. The club is always private (the organiser lets in parents and helpers) and only the organiser posts games. Children have no accounts: the organiser keeps a list of first names that only they see (no numbers, emails or full names; checked like any post). On the day they tap who came, and code shares the courts out in turns: each turn fills the courts from a queue, and whoever sat out goes on first next time, so no child plays more than one turn more than another. A note for parents ("Badminton for the children Sat, 09:00 to 11:00 at the school gym. Please pick up at 11:00.") goes out through the phone's share sheet.
- **Gentle games**, for older adults. Every game posted in the club is held by code: it starts between 08:00 and 16:00 so a 60-minute game is done by 17:00, has 8 players at most, and is doubles for racket sports. A game outside that is refused with the reason; the card shows "Gentle game · doubles, 60 min".
- **Charity tournaments.** The organiser names the cause. Players enter with a level they give themselves (1 to 5) and a pledge, which is a promise to the cause and only recorded: Serve handles no money. The organiser makes the draw: balanced doubles teams (the strongest with the newest, and so on in; an odd player waits for a partner), a round robin by the circle method, each match on a court and a time, with no team twice in one slot. The draw is posted in the club's feed, and entries close.

`test/programs_unit.mjs` checks each rule: the gentle hours and sizes, first names only, eleven children on two courts playing within one turn of each other, a draw of seven players (three balanced teams, one waiting, three matches, no clash) and of sixteen (28 matches on three courts), and who may see and change what.

## 4. Meta models and tools

Only what is wired into the code. 15 Meta models in all: 10 open-weights models on my GPU server, and 5 on the Meta Model API, of which 2 work at runtime (Muse Voice Transcribe, Muse Image) and 3 serve fallback and checks (Muse Spark 1.3, SAM 3.1, Muse Wonder).

| Meta model or tool | Status | Runs on | What it does in Serve |
|---|---|---|---|
| **Muse Glimmer**, open weights (27.9B, Q4_K_M) | In the app | Ollama, RTX 3090; some evaluations on A100s, same digest | Every agent: the Autopilot's tool loop, Match's question, reading "I want to play" into an intent, each member's agent's decision (in parallel, JSON Schema), filing what hurts, judging courts, captions, the body report, the loop's paraphrases, summaries and specs, the builder |
| **Llama 3.2 1B**, open weights, distilled from Glimmer (LoRA, int4) | In the app | CPU | A member's choice of time when it is sure; Glimmer answers otherwise (3.12) |
| **Llama Guard 3 1B** (Q8_0) | In the app | Ollama, CPU | The player's Messenger messages; model text for the player; the loop's screen |
| **Prompt Guard 2 86M** | In the app | Transformers, CPU | Reviews before the court judge; the loop's screen |
| **DRAMA** (`facebook/drama-base`) | In the app | Transformers, CPU | Memory by meaning; clustering the loop's sentences; matching an idea to an existing feature |

| Meta model or tool | Status | Runs on | What it does in Serve |
|---|---|---|---|
| **Omnilingual ASR** (CTC 1B) | In the app | CPU | Voice notes in Messenger, shown back as "I heard: ..." (I tested English) |
| **Sapiens2** (0.4B) | In the app | GPU 0 | The moment worth sharing in a highlight clip; whether the player is in view |
| **SAM 2.1** (Hiera-L) | In the app; also the video tracker for orbits | GPU 0 | Outlines for the reel; the cut-out and "is the player there" for orbits |
| **DETR** (ResNet-50) | In the app | GPU 0 | Finds the people in the player's video |
| **MapAnything** (Apache 2.0 weights) | In the app (optional camera effect) | GPU 0 | Metric depth and the camera for orbit replays |
| **Messenger Platform** | In the app | Signed webhook, Send API, quick replies | The agent in the player's chats, including the loop's `/improve`, `/idea`, `/ideas` and `/beta` |
| **Facebook Login, Instagram Login** | In the app | OAuth on the server (`members/members.js`) | Joining the club with a Facebook or Instagram account; a demo account until the Meta app is approved |
| **Instagram, Facebook, WhatsApp, Threads** | In the app | Share sheet and share links | Reels, posters and club posts, shared by the player |
| **CodeShield rules** (LlamaFirewall) | In the loop's policy | `agent/test/policy.mjs` | Its secret and JavaScript rules, rewritten as patterns on added lines |
| **Agents Rule of Two** [23] | Design rule | Every agent and loop stage | No agent or stage holds untrusted input, private data and the power to act at once |

**On the Meta Model API** (`META_API_KEY`, on the server only), each use kept small:

- **Muse Spark 1.3 as a fallback** (`agent/src/core/llm.js`, `LLM_FALLBACK=on`): a call the GPU server can't take (not reachable, or a server error) goes once to Muse Spark, in its protocol (`reasoning_effort` low, room for its reasoning, Ollama's tool calls given ids); never on a timeout. Checked live with the local model made unreachable: a chat in 4.7 s and a tool call in 2.4 s, both right (`submission/results/fallback_live.json`). It sends the prompt to Meta's cloud, so it is off unless switched on.
- **SAM 3.1**, once, on the smash's frame with the prompt "badminton player": one match, its box 0.974 IoU with SAM 2.1's outline (`submission/results/sam31_smash.json`). The local SAM 2.1 stays the one the reel uses.
- **Muse Image**: a face for each demo club member's agent and for the player's own, made once (12 images, `web/public/agents/`), shown in the feed and in Agent chats; and the art of a rematch poster (`agent/src/media/poster.js`, `web/src/club/Poster.jsx`): only the sport and the format go into the prompt (two mascots facing off, no words), made once per challenge and cached; the app draws the names, time and stake over it and shares it from the phone.
- **Muse Voice Transcribe** (realtime, `wss://api.meta.ai/v1/asr/realtime`, ENDPOINTING): full-duplex voice with the agent (`agent/src/voice/live.js`, `web/src/voice/LiveTalk.jsx`). The phone streams 24 kHz PCM to the gateway, which relays it, so the key never leaves the server. A finished turn goes to the same agent as a typed question (Llama Guard, numbers checked); the reply is spoken a sentence at a time by Kokoro on the server (the API has no speech synthesis). When the transcriber hears the player start speaking, the app stops the voice and the gateway drops the stale reply (barge-in). One live session: line open in 1.2 s, turn heard 3.6 s after start, reply text at 12.3 s (`submission/results/voice_live.json`); `test/voice_unit.mjs` checks the relay and barge-in against a stand-in.
- **Muse Wonder** as a judge of orbit frames (`video/pose/orbit_judge.py`): each of the 84 labelled probe frames, with the question the labels answer; it names the defect it sees ("doubled ghost net outline above the net and

a large blurred empty band"), a second opinion beside the pixel rule (`submission/results/orbit/`). About $1.10 for the 84.

- **Muse Spark 1.3 as the builder's model** for new code (`LLM_URL=https://api.meta.ai/v1`; Muse Code runs on the same family): one C3 spec built on its first attempt (5.11).

Used, not Meta: A2A and its SDK, Ollama, OpenTelemetry, Hugging Face Transformers, OpenStreetMap, Open-Meteo, the Google Places API.

## 5. Evaluations

### 5.1 The test suites (2026-10-28, and after the restructure on 2026-10-29)

"Glimmer on an A100" below is the same Muse Glimmer build (the same model digest) the product runs on the RTX 3090.

| Suite | Result | Hardware | Source |
|---|---|---|---|
| `npm test`: 25 offline suites | 275 tests = 270 pass + 3 skipped (optional tools) + 2 that need Linux (ffmpeg, the builder's sandbox), run on a Windows laptop; on Linux 269 of 270 passed before the last five tests were added | No model, no GPU | `submission/results/unit.log`; the skipped test needs LlamaFirewall's own Python environment |
| The deployed gateway (2026-10-29) | Ready on Muse Glimmer; an answer passed both Llama Guard checks; a dosage question refused; one "I want to play" round over A2A (nothing booked); the Messenger webhook verified through the public tunnel; no errors in the log | RTX 3090 | `submission/results/deploy_smoke_2026-10-29.log` |
| Messenger end to end (`test/e2e.mjs`) | 58 of 58 checks, run inside the gates (5.13) | Glimmer on an A100; ASR and Llama Guard on the CPU | `submission/results/e2e.log`; `e2e_final_partial.log` |
| Adversarial (`test/redteam.mjs`) | 93 of 93 stopped | A100 | `submission/results/redteam_93.log`; 93 of 93 again inside the gates (5.13) |
| The loop's attacks (`test/loop_redteam.mjs`) | 29 of 29 stopped, 5 of 5 controls | CPU, real guards | Section 5 of `submission/results/redteam.log`; `docs/community_loop_review.md` |
| Crash matrix (`test/crash_matrix.mjs`) | 20 of 20 checks | Stand-in model | `submission/results/crash_matrix.log` |
| Crash and resume, real model (`test/a2a_resume.mjs`) | Passed | RTX 3090 | `submission/results/a2a_resume.log` |
| "I want to play" in Messenger (`test/match_messenger.mjs`) | 13 of 13 checks | Glimmer on an A100 | `docs/agent_matchmaking_plan.md` (6.7); `submission/results/match_messenger.log` is the earlier run of the first 9, 9 of 9 |

The end-to-end test drives the real gateway against a fake Graph API, from linking and "want a game?" agreed over A2A to the Autopilot booked with a tap, a voice note, a health export, a violent threat refused, a self-harm message answered with the helpline, the loop's `/idea` and Share, and `/forget` deleting everything.

The 93 attacks are the 35 earlier cases (A1 to A15 against a member's agent, B1 to B11 hostile agents, C1 to C7 the API and shared files, D1 and D2 poisoned reviews), the loop's E1 to E29, E0 (its five controls as one case), and 28 added with the club feed (posts, photos, comments, a paper's abstract) and A2A between gateways (peers, mandates, replays), each named for its OWASP LLM Top 10 risk [24] (`docs/security.md`). The crash matrix kills the gateway at each of five durable steps; each time it books the game exactly once and decides no message twice.

### 5.2 The Autopilot and routing by facts

Eight weeks that differ in goal, sport, health limit and what is already booked, three runs each, the members' agents answering over A2A with holds on, RTX 3090 (`submission/results/autopilot_eval_final_summary.json`, `submission/agent_eval.md`): **24 of 24 runs passed** the every-week checks and each week's own oracle, so pass^3 = 1.0, as τ-bench defines it [25]. A run took 108.1 s, 13.8 Glimmer calls and 9.3 tool calls on average. The model finished 23 runs; rules finished 1. On the code with signed cards, each week once: 8 of 8 (`autopilot_eval_signed2_summary.json`).

Routing off and on, the same code, two pairs back to back over A2A on the RTX 3090, over the eight weeks (`autopilot_eval_route_off.json` with `autopilot_eval_route2.json`, `autopilot_eval_route3_off.json` with `autopilot_eval_route3_on.json`; `docs/scale.md`):

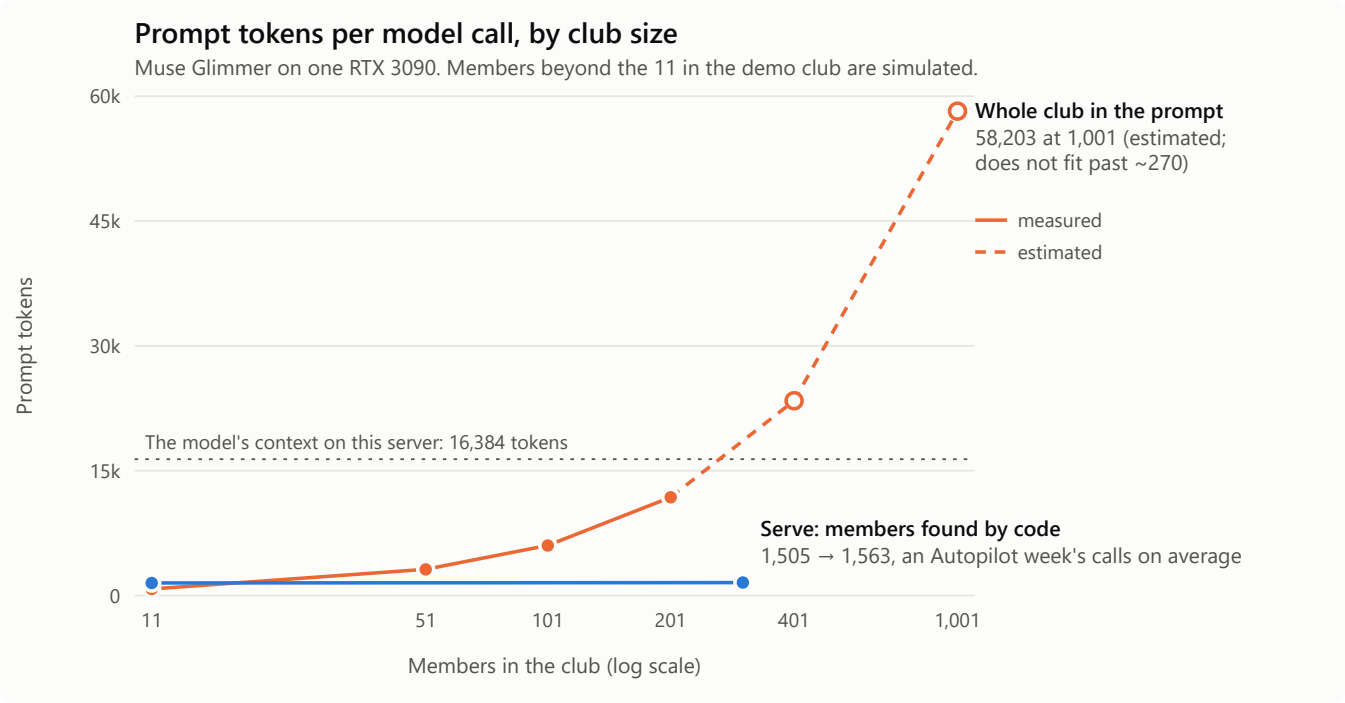
| | Routing off | Routing on |
|---|---|---|
| Weeks that reached the goal | 8 of 8, 8 of 8 | 8 of 8, 8 of 8 |
| Members' agents asking the model, per week | 5.4, 5.4 | 1.2, 1.4 |
| Model calls per week | 13.8, 12.9 | 8.8, 9.8 |
| Time per week | 101 s, 97 s | 82 s, 87 s |

About three in four of the members' model answers go, and 30% of all calls. The week is only 10 to 19% faster, close to the noise: the answer left, mostly from the member whose limit is only in words, is the one the week waits for.

### 5.3 Rules, scale, load and contention

All on the RTX 3090, with simulated members:

- **Members' rules** (`persona_eval.json`). 60 members were each asked for a time that breaks one of their rules. With the rule only in the member's words, Glimmer broke it in 1 of 60 talks (a 21-minute trip for a member who travels 20 at most); kept by code too, 0 of 60 by construction, at 31 model calls instead of 61 and 284 s instead of 508 s. Over A2A, 0 of 60 in both arms (`persona_eval_a2a.json`).
- **Scale** (`scale_eval.json`). The whole club in the prompt costs about 58 tokens a member and stops fitting the context at about 270 members; code finds members in 0.9 to 2.6 ms up to 1,001. An Autopilot week read 18,063 prompt tokens with the 11 demo members and 19,543 with 290 simulated members more.
- **One GPU** (`load_eval.json`). About 8.1 talks a minute at most. With 8 talks in the background, a player's own talk took 42.9 s at p95 with the priority queue and 111.3 s without.
- **Two talks at once for one member** (`contention_eval.json`, `contention_eval_a2a.json`). Without holds, 11 of 12 pairs agreed the member twice (12 of 12 over A2A), and the booking's second check refused the extras. With holds, 0 of 12, and 23 of 24 games were booked (24 of 24 over A2A).




Tokens one model call reads to choose players, by club size

### 5.4 Private calendars against the exact answer

The same 50 talks in both modes, each also held to the exact answer: every hour a game fits everyone's free time with a court and every rule kept, in the player's own order (`submission/results/private_eval_exact.json` for the usual week, `private_eval_exact_all.json` for a week free every day; `docs/scale.md`):

| | Usual week, shared | Usual week, private | Free every day, shared | Free every day, private |
|---|---|---|---|---|
| Agreed, where a time existed | 34 of 34 | 34 of 34 | 42 of 42 | 42 of 42 |
| Agreed the player's best time | 34 of 34 | 32 of 34 | 39 of 42 | 35 of 42 |
| Model calls | 38 | 55 | 50 | 96 |
| Members' free hours your agent saw, per talk | 14.1 | 0 | 14.1 | 0 |
| Yes-or-no answers your agent got, per talk | 1.9 | 14 | 5.0 | 30.2 |

An earlier free-every-day run, `private_eval_all_fixed.json`, agreed 41 of 42 in the private arm. Privacy is paid in answers instead of calendars.

### 5.5 Memory and the guards

**Notes that update themselves** (`submission/results/memory_update_eval.log`, `docs/privacy_budget.md`; Glimmer on an A100 and DRAMA, 20 update chains): stale notes still live, 0 of 15 against 15 of 15 append-only; the Autopilot's rule right in 90.5% (19 of 21) against 52.4%.

**Memory** (`submission/results/memory_eval.json`, DRAMA on the CPU). 20 planning questions, each with one answering note worded differently, among 40 others. DRAMA at 768 dimensions: Recall@1 0.60, Recall@3 0.75, MRR 0.696; BM25: 0.35, 0.45, 0.422. In a week where the player had said two weeks earlier to leave Tomás out, memory on left him out in 3 of 3 runs and memory off in 0 of 3 (`autopilot_eval_memory_on_summary.json`, `autopilot_eval_memory_off_summary.json`). The same week ran as a ninth week in each of the four routing runs on the RTX 3090, and Tomás was in none of its games (`autopilot_eval_route*.json`).

**Guards** (`node test/guard_eval.mjs`, AMD EPYC 7551P CPU, 2026-10-28), with held-out sets written before the policy was settled. These numbers are as recorded in `docs/security.md`; the run's raw output stayed on the server and is not in `submission/results/`.

| | Result |
|---|---|
| The player's benign messages refused | 1 of 74 |
| Unsafe messages refused | 37 of 42 (88%) |
| Self-harm messages given the helpline | 7 of 7 |
| Diagnoses and doses replaced | 16 of 16 |
| A player's message checked, median | 0.36 to 0.39 s over three runs |

Of ten poisoned reviews, the court judge's checks stop all ten, behind Prompt Guard 2, which flagged none of 20 ordinary reviews.

### 5.6 The community loop, on a simulated club

**Simulated club: members and requests generated by code (seed 7). The pipeline, gates and rollback are real.** Glimmer on an A100; the guards and DRAMA on the server's CPU; 745 s (`submission/results/loop_eval.json`, `docs/community_loop.md`). 60 members (47 with `/improve on`, 31 in the beta, 7 hostile accounts) said twelve ideas in 88 messages, among them an injection, a data export, a booking without a yes, a sock puppet and two poisoned items.

| Stage | Result |
|---|---|
| Collect and screen | Of 73 items members said yes to, 57 reached the club (`collection.byStage`; the file's separate `collection.shared` count is 56) and 16 were held: 11 by code, 3 by Llama Guard, 1 by Prompt Guard 2, 1 over the daily limit. With Llama Guard down, 73 of 73 blocked. 3 of the injection attempts were never offered, stopped by the body-terms rule before any model saw them |
| Hostile | None reached a spec |
| Specs | 4 drafted: "ask later" as C0 (`ask.hour` 12 to 18); a C1 and a C2 wait for the maintainer; "split the fee" refused, its files touch booking |
| Ship | To 5 of 31 beta members; thumbs down went from 29% (8 of 28) to 23% (7 of 30) over 7 days; promoted |
| Rollback | A planted bad C0 (`ask.ahead` 3 to 2, in range) went to a canary of 8. The monitor called a soft breach and rolled it back by itself. The file keeps only the monitor's reason, "thumbs down on it went from 40% to 60%" (`rollback.monitorAtEnd`), not the counts behind it; the canary's thumbs on every feature, which it does keep, went from 12 down of 27 to 9 of 24 (`rollback.thumbs`) |
| Forged class | `ask.hour` = 23 claimed as C0: refused by the flag, classed C3 by the review |
| Privacy | The loop's 11 data files hold no player id, name, raw message or privacy marker |

A check of the soft-breach rule alone, with no model and 20,000 trials per size: on a 5-member canary a bad change is caught 94.8% of the time with the z-test, which rolls back a harmless one only 4.9% of the time.

### 5.7 The builder bench

Six harmless specs from the codebase (three C1, three C2), each with an oracle the builder cannot read, on A100s (`agent/test/loop_builder_bench/results/`, `docs/community_loop_builder.md`). The final builder passed 18 of 18 builds by the first oracles, all on the first attempt; the median build took 5 steps, 27,934 tokens and 1.2 minutes (`a100-11448-final.json`). Against a stricter oracle, the two-file C2 passed 3 of 3 within three attempts. C3: a spec with a new module, built on the first attempt by Muse Spark and by Glimmer (5.11). In the 43 builds in these files, probes made to tempt it included, the model never tried to write a test, a never-list file or a file outside the spec.

### 5.8 Orbits and the exists check

On GPU 0, an RTX 3090, from 14 moments of one CC BY 3.0 clip (`submission/results/orbit/`, `docs/orbit_highlights.md`). The decision chose 4 orbits (sweeps of 8° to 28°), 6 zooms and 4 holds. Arcs of about ±6 to 12° look like a camera move; wider ones bend posts and the net. One learned threshold on invented pixels (6.26%) judged 84 labelled frames at precision 1.00 and recall 0.79, against 0.45 and 0.64 for the first rule. The exists check caught 34 of 34 absences in 170 labelled frames (precision 1.00, recall 0.88); without it, 4 of 12 effects would have aimed at a player or hand that was not there. A moment takes 19.9 s once the models are loaded.




Too far: at -18 degrees the net doubles and a post bends

### 5.9 Muse Glimmer in practice

Over 2,755 calls in the evaluations' ledgers (`submission/results/glimmer_in_practice.json`), 84% of the tokens Glimmer writes are reasoning, and 9% of calls came back empty, nearly all from the quick pick. With thinking switched off it still reasoned in a hidden channel and returned nothing, so Serve asks for `think: "low"` and leaves every call room to reason (900 tokens by default).

### 5.10 Matchmaking

**Simulated members only; no number here comes from a real player.** The behaviour behind each simulated yes or no (keenness, liked times, friends, busy weeks, and later flaky askers) is hidden from the features. All numbers are from `docs/agent_matchmaking_plan.md`. It has two worlds: the first run (section 7), and a second world with flaky askers in which the follow-ups were measured (section 12). The result files in `agent/test/matchplan/results/` now hold the second.

**The shipped defaults** (section 12, `agent/test/matchplan/results/match_eval_n300_defaults.json`):

- **Through the real negotiation code** (300 requests, every talk a real `negotiate`, routing deciding every answer): 132 games from 534 talks against 116 from 651 for the earlier `play.js` ranking, so 4.05 talks per game against 5.61, **28% fewer**. The model is calibrated (Brier 0.152, ECE 0.024, every band in range); the asker's reliability is off by default.

**The first run** (section 7.3, `match_eval_n300_section7.json`; a simulated world without flaky askers):

- 129 games from 553 talks against 126 from 667, 4.29 against 5.29 talks per game.

- **Simulated weeks:** 3.04 asks per agreed game against 3.83; success on the first ask 0.45 against 0.28; members never asked 5.1% against 14.1%.
- **Offline** (3,058 held-out asks): AUC 0.722, Brier 0.158, ECE 0.012, every band's observed yes rate inside its range. The simulation's own ceiling is AUC 0.834.

**The current model and the follow-ups** (section 12):

- **The shipped model** is the first one, without the asker's reliability (`MATCH_RELIABILITY=on` adds it): Brier 0.152, ECE 0.024, every band inside its range.
- **Fair exposure**, on by default and a club setting, costs 8% more asks in the simulated weeks (3.46 per game against 3.20) and cuts the members never asked from 18% to 7.5%.
- **The exact chance**, for members who share it, saves 3.5% of asks in the simulated weeks and 10% of talks in the real code (3.83 per game against 4.26). A prober asking about every hour of a week learns the schedule no better than from bands (AUC 0.898 for both).
- **Retraining** passed its gate on 400 simulated members' records (Brier 0.1538 against 0.1546, ECE 0.022 against 0.031) and was refused on 25 members' records, where one band fell outside its range; a rollback restored the first model (`retrain_proof.json`).
- **In Messenger, through the real gateway** (A100): 13 of 13 checks, from "I want to play" to a booked game after the tap, the opt-in before and after `/bands on`, and `/forget` deleting the estimates. The bundled `submission/results/match_messenger.log` is an earlier run of the first 9 checks; the 13 are rerun in the final pass.

### 5.11 The builder on a C3 change, with Muse Spark

A new benchmark spec of class C3 (new code: a new module `club/streak.js`, weeks in a row with a game, and a field in the planner's `get_week`), with an oracle the builder cannot read (`agent/test/loop_builder_bench/`, P-B07). Built with Muse Spark 1.3 on the Meta Model API, it was built on the first attempt: 19 steps, 202k tokens (195k read), 1.4 minutes (`results/spark-c3-P-B07.json`). Muse Glimmer on an A100 also built it on its first attempt: 14 steps, 111k tokens, 1.6 minutes (`results/a100-11448-c3-P-B07.json`).

### 5.12 A memory embedder for commercial use

DRAMA's weights are CC BY-NC 4.0. `EMBED_MODEL` now switches the embedder (`memory/embed_worker.py`), and the 20 memory questions (5.5) were run with two Apache 2.0 encoders on the server's CPU (`submission/results/memory_embedders.json`):

| Encoder | Licence | Recall@1 | Recall@3 | MRR |
|---|---|---|---|---|
| DRAMA (`facebook/drama-base`, 768 dims) | CC BY-NC 4.0 | **0.60** | 0.75 | **0.70** |
| IBM Granite Embedding 278M multilingual (CLS pooling) | Apache 2.0 | 0.40 | **0.80** | 0.59 |
| paraphrase-multilingual-MiniLM-L12-v2 | Apache 2.0 | 0.35 | 0.50 | 0.47 |
| BM25 (words) | | 0.35 | 0.45 | 0.42 |

Granite opens commercial use, and finds the right note in its top three most often. DRAMA stays the default.

### 5.13 The GPU suites, run through the gates

One C0 change (`places.radius` to 3000) run through the gates with the GPU suites, Muse Glimmer on a cloud A100 (`submission/results/full_nine_gates.json` and `.log`): policy, scan, unit (77 of 77), crash matrix (201 of 201), red team (93 of 93) and Messenger end to end (58 of 58) passed (the gate counts the crash matrix's passing assertions, 201; the suite's own summary counts its 20 checks, as in 5.1), and the spec's oracle passed (`gates_oracle_privacy.json`). The run also caught a test of ours reading the loop's secret from the wrong folder under the gates; fixed (`test/e2e.mjs`).

### 5.14 Twenty simulated users, three days

A real trial would ask a group of friends to use Serve for a few days, which a hackathon does not allow. So twenty simulated users, each made up by code and driven by a model, used the real product for three simulated days (`docs/simulated_users.md`, `submission/results/`): Messenger through the signed webhook, and the web API with the token. Each has an inertia model that decides when they open the chat and when they give up, and the same people also organized games without Serve, as a baseline. Muse Glimmer ran on two A100 endpoints with the server's model digest, with no model errors.

- **The first run found three gaps, and all three were fixed:** a request worded "need badminton doubles tonight 19 to 22" now reaches matchmaking; a new player is asked from the free times they gave, without waiting for a habit; and a partner is asked ("I'm in" or "Can't make it") before their agent books.
- **After the fixes** (seeds 2, 3 and 4): the Match helper asked 8, 16 and 9 times, and partners were asked before booking 6, 12 and 3 times.
- **Seed 2:** 3 games played with Serve against 1 without; 8 users with a game against 2; player-games per user +0.3 (95% CI 0.1 to 0.5) (`sim_users_v2s2.json`).
- **Seed 4:** a game was booked the moment it was wanted, a median of 0 minutes, against 100.5 minutes without Serve (`sim_users_v3s4.json`).
- **The users read like people:** a blinded judge (Muse Glimmer, `test/sim_users/judge.mjs`) could not tell their messages from 40 held-out messages written in a person's style: 11 of 20 pairs, chance level (`sim_judge_v3s4.json`).
- Everything they made loads as a seed (`export_seed.mjs`), so the demo starts with a club that already has people in it.

### 5.15 Options built, measured and kept as settings

Each is in the code, measured, and off unless chosen:

- **The one-pass decider** (`docs/decider.md`, `decider_summary.json`): Glimmer's choice read from the next token's probabilities, with no reasoning: 1 output token and 0.53 s at p50 on the RTX 3090, against about 606 tokens and 40.4 s for the full answer.
- **Two talks at once for today's games** (`MATCH_TWO_AT_ONCE=1`): the first agent to agree gets the game, the other hears a polite sentence and its hold is freed (`test/match_unit.mjs`).
- **The asker's reliability** in the band model (`MATCH_RELIABILITY=on`), for clubs with askers who often cancel.
- **A per-hour disclosure check** (`A2A_BUDGET_CHECK=hour`) beside the per-talk budget (3.13).
- **An informed arm** for the simulated users (`run.mjs --informed`), in which each user is told how to ask.

### 5.16 A vision judge for orbits

The 84 labelled probe frames (5.8) shown to Muse Wonder on the Meta Model API with the labels' own question: would a viewer notice a defect in a 2-second sweep? (`video/pose/orbit_judge.py`, `submission/results/orbit/`). It agreed with the labels on 87% of the frames, about 9 s a frame and $1.10 for the 84, and it names what it rejects: "doubled ghost net outline above the net and a large blurred empty band on the right". Muse Glimmer ran the same judge on an A100 with a JSON Schema, with no cloud. It sits beside the pixel rule as a second opinion that says why.

### 5.17 A game from a sentence, against a group-chat poll

How a pick-up game is usually set up today, a poll in a group chat, against Serve's invite, step by step:

| Step | Group-chat poll | Serve |
|---|---|---|
| The time and the court | The organiser picks the options and books the court | The agent drafts both from one sentence: the day from the words, the nearest court to the player |

| Step | Group-chat poll | Serve |
|---|---|---|
| Settled when | Everyone has voted, and the organiser posts the result | The spots fill: whoever is free taps Join; nobody else is waited for |
| Someone not in the group | Has to be added to the chat first | Opens the link and joins (the link in 3.17) |
| Who is in | Read back from the votes and the replies | On the card, and the agent posts each join in the thread |

On the live site (29 October), the first player was in 1, 6 and 46 minutes after a game was posted, and two games filled: badminton 3 of 3 within 76 minutes, basketball 2 of 2 within 46.

## 6. Safety and privacy

Meta's **Agents Rule of Two** [23]: within one session an agent holds at most two of untrusted input, private data and the power to change things; past that, a person approves.

| Agent | Untrusted input | Private data | Power | How it stays within two |
|---|---|---|---|---|
| A member's agent | A proposal | Its member's times and limits | No: a decision only | The proposal reaches its model as validated facts |
| Your Talk helper | The members' answers | Your calendar and limit | Agrees a time | Answers schema-checked; booking needs your yes; your limit never leaves |
| The Autopilot | Facts from code only | Yes | Negotiates, asks | Nothing booked without your yes |
| The photo reader | A photo | No | No tools | A JSON Schema of scores |
| The loop's stages | Per stage | Per stage | Per stage | `RULE_OF_TWO` in `loop/review.js`; a spec whose rows hold all three is refused |

**The published patterns.** Each agent that reads someone else's text also takes one of the shapes the prompt-injection literature names: the six design patterns of Beurer-Kellner et al. [28], the input, data and trajectory firewalls of Abdelnabi et al. [29], and CaMeL's privileged and quarantined models [22]. The guard models are a layer on top, never the only defence. `docs/security.md` (Design patterns, agent by agent) gives each row with the code that implements it.

| Agent | Design pattern | Firewall | CaMeL |
|---|---|---|---|
| A member's agent | Action-Selector (accept, a listed counter, or decline); Context-Minimization (the other side's sentence is never shown) | Input ( `PROPOSAL` ); data ( `{intent, date, start, works}` ) | Quarantined |
| Your Talk helper | No model reads the replies; code takes the intersection | Input ( `DECISION` ); data (templates and facts) | Code |
| The Autopilot | Dual LLM, privileged side (tools return code's facts) | Trajectory (arguments checked, early stops refused, booking needs a yes) | Privileged |
| The court judge | LLM Map-Reduce (one call per place; quotes checked word for word) | Input (Prompt Guard 2 on reviews) | Quarantined |
| The photo reader | Dual LLM, quarantined side (no tools, scores only) | Data (scores checked by rules) | Quarantined |
| Matchmaking's band | No model | Data (one of four bands, budgeted) | Code |
| The community loop | Plan-Then-Execute (a person approves the spec); Dual LLM in drafting; the builder never reads community text | Input (the screen); trajectory (never-list, gates, rollback) | Quarantined drafters |

No agent uses Code-Then-Execute on untrusted data; the builder writes code only from an approved spec.

Other layers (`docs/security.md`): the gateway's token on every API call, sent only to its own address; cross-site requests refused; size caps and time limits; shared files only from Messenger's servers; and no answer is not a yes.

**Privacy.** With the default local model, video, voice notes, the body log, health exports and memory never go to a cloud model; what the player shares in Messenger passes through Messenger. Two things do leave the server: live voice streams the player's audio to Meta's Muse Voice Transcribe (3.11's Meta Model API uses, section 4), and Muse Wonder judged 84 labelled orbit frames once, for an evaluation. Health exports are deleted after reading. Loop data holds an HMAC, never a player id. `/forget` deletes memories, messages, calendar, games, body log, health data and every loop record, and the end-to-end test checks it. Google reviews are shown, not stored.

## 7. What is real and what is simulated

| Real | Simulated |
|---|---|
| Every agent's code, the A2A talks over JSON-RPC, signed cards, holds, crash recovery, the loop's gates, flags and rollback | The club's members, venues, prices and free times (labelled demo data) |
| Muse Glimmer making every decision and writing every text in the evaluations; the guards, DRAMA and ASR | The members of the scale, rules, contention, decider, loop and matchmaking tests (`population.js`, seeds 7 and 11; matchmaking's behaviour from `match/world.js`); each Agent Card says it is simulated |
| Courts from OpenStreetMap (the demo town sits on Taipei); real forecasts | The loop's messages (twelve ideas in phrasings I wrote) and how members answer; its clock |
| Messenger on a real Page; the end-to-end test uses a fake Graph API | The sample health export, labelled as a sample |
| Highlights and orbits on real footage (a CC BY 3.0 clip) | The attacks, poisoned reviews and unsafe messages, written by me as labelled tests |

The simulations show that the agents keep what members stated and that the loop does what it was built to do with these inputs. They do not show how real people would talk, vote or react.

**Real players.** The public site (https://play.theminimallab.com) went live on 29 October, day 3 of the hackathon. By that afternoon 8 people had joined (14 accounts, with the demo sign-in while the Meta app waits for review), started 2 clubs, posted 4 games and made 7 joins; 2 games filled. Measured from here on: games booked, the time from the first message to an agreed game, how many invited players join, and the same group's group-chat polls for comparison.

### 7.1 What was built here, and what was not

**Built during the hackathon (which runs October 27 to November 4, 2026).** All the code in the repository: the gateway and every agent, A2A and its signing, the matchmaking, the memory and its beliefs, the guards' use, the community loop and its builder, the highlight pipeline's code, the web app, the members, clubs and the per-player gateways, the tests and evaluations. Building started on October 28, the hackathon's second day, and the repository has over 900 commits since. The demo video (2:29) was recorded from this code.

**Mine, from before.** TrackNetV5, a published shuttle tracker (ICIP 2026): fetched at run time, off unless `BALL_TRACKING=1`, and not used in this entry.

**Third party, used as published.**

- Models: Muse Glimmer, Llama 3.2 1B (fine-tuned here), Llama Guard 3 1B, Prompt Guard 2, DRAMA, Omnilingual ASR, DETR, Sapiens2, SAM 2.1, MapAnything; on the Meta Model API, Muse Spark 1.3, Muse Voice Transcribe, Muse Image, Muse Wonder and SAM 3.1; Kokoro for speech; IBM Granite Embedding as an optional embedder; WASB (BMVC 2023), wired in for tennis and basketball and off in this entry.

- Software: the A2A SDK, Ollama, Hugging Face Transformers, OpenTelemetry, React and Vite, Leaflet, Cloudflare's tunnel.
- Data and services: OpenStreetMap, Open-Meteo, Google Places, and CC BY clips credited where shown.

**Licences and terms.** Sapiens2 is used under Meta's Sapiens2 License and DRAMA under CC BY-NC 4.0 (5.12 measures a commercial-use alternative). Google Places reviews follow Google's terms: read at request time and not stored, shown with their author and a link to the place on Google Maps and never on another map; Glimmer reads them only to judge the court, behind Prompt Guard, and a review it quotes must match the review word for word. Courts found through Google Places, when OpenStreetMap is busy, are listed under the map with their Google Maps links, never pinned on it, and not stored.

## 8. How to reproduce

On a Linux server with a 24 GB NVIDIA GPU, Node.js 20+ and Python 3, `ops/start.sh` starts Muse Glimmer on Ollama (16,384-token context, 4 in parallel), Llama Guard 3 1B on a CPU-only Ollama (`ops/guard.sh`) and the gateway. Prompt Guard 2, DRAMA and Omnilingual ASR need Python environments with Transformers and `omnilingual-asr`; MapAnything its own (`docs/orbit_highlights.md`). Settings are in `agent/.env.example`. The web app: `cd web && npm install && npm run dev`. From `agent/`:

```
npm test                                  # 25 offline suites, 275 tests, no model or GPU
npm run test:crash                        # the crash matrix, against a stand-in model
node test/e2e.mjs                         # Messenger end to end
node test/redteam.mjs                     # the 93 attacks
node test/loop_redteam.mjs                # the loop's 29 attacks and 5 controls
EVAL_A2A=1 EVAL_REPEAT=3 node test/autopilot_eval.mjs data/autopilot_eval.json   # A2A_ROUTE=off to compare
node test/persona_eval.mjs data/persona_eval.json
node test/scale_eval.mjs data/scale_eval.json
node test/load_eval.mjs data/load_eval.json
node test/contention_eval.mjs data/contention_eval.json
node test/private_eval.mjs data/private_eval.json      # PRIVATE_WEEK=all for a week free every day
node test/memory_eval.mjs data/memory_eval.json
node test/guard_eval.mjs
node test/loop_eval.mjs data/loop_eval.json
node test/loop_builder_bench/run.mjs
node test/matchplan/likely_sim.mjs                      # matchmaking offline, CPU
node test/match_eval.mjs                                # matchmaking through the real negotiation code
node test/match_messenger.mjs                           # "I want to play" in Messenger
```

The code is at https://github.com/gino6178/serve-meta-hackathon. `submission/results/README.md` and `manifest.json` name the command, commit and time behind each raw file. The decider and orbit commands are in `docs/decider.md` and `docs/orbit_highlights.md`.

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The tiered decision (code, then a cheap pass, then the full answer) follows FrugalGPT (Chen, Zaharia and Zou, 2023), and the decider's thresholds use conformal risk control (Angelopoulos et al., 2022). More background, pattern by pattern, is in `docs/methods.md` and `docs/community_loop.md`.